



Fracture toughness and crack propagation in LX-17 and PBX 9502 insensitive high explosives

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ABSTRACT

High explosive materials, and in particular insensitive high explosive (IHE) materials, are of significant interest. TATB-based LX-17 and PBX 9502 are two IHE's and mechanical properties are an important metric of performance. In this work we present, for the first time, a study of fracture as a function of temperature and displacement rate and starter crack length on billet pressed IHE materials using a standard three point bend technique. We demonstrated that the technique can yield consistent results over time by measuring samples from the same lots 4 years later and observed that the fracture toughness of LX-17 and PBX 9502 are similar. We also determined the temperature dependence for different starter crack lengths and found that the fracture toughness generally decreases with larger starter crack sizes and temperatures. For the LX-17, where a wide temperature range of -60 C to 50 C was investigated, the impact of temperature on fracture toughness between -20 C and 23 C was less than expected. For the PBX 9502 we observed the difference in fracture toughness values between -20 C and 23 C did depend on the particular lot, and also there was a tendency of the fracture toughness to scale with the tensile strength. We also investigated the effect of a lower displacement rate and observed that while the impact on the fracture toughness was relatively small, the samples surprisingly did not break (fail) after exceeding the peak load. We took advantage of this to perform x-ray computed tomography (CT) of the samples to image crack propagation in the intact samples. We observed that cracks were generally straight and unaffected by the microstructure in the LX-17, in contrast to the PBX 9502 where the direction of the cracks would frequently change and were likely influenced by the underlying microstructure.

1. Introduction

High explosive materials are of significant interest for a number of applications including mining, welding, construction/demolition, and defense (Meyers and Shanley, 1990; Drell, 2007; Linse and Lalwancy, 1984; Blanchard, 2002; McKenzie et al., 2019; Cook, 2016; National Nuclear Security Administration, 2018; Mathieu and Stucki, 2004; National Nuclear Security Administration, 2020). Within the defense realm they are not only used for traditional armaments (Mathieu and Stucki, 2004) but are also a critical component of the nuclear deterrence enterprise (Drell, 2007; Cook, 2016; National Nuclear Security Administration, 2018; National Nuclear Security Administration, 2020). For nuclear deterrence applications a subset of high explosives materials termed insensitive high explosives (IHE) are highly desired; these IHE's can be developed with a significant safety margin so that the probability of an accidental detonation is extremely small (Drell, 2007; Cook, 2016; National Nuclear Security Administration, 2018; National Nuclear

Security Administration, 2020). It has been said that there is no single greater improvement in weapons safety than moving to IHE (Cook, 2016; National Nuclear Security Administration, 2018). Two leading IHE's are the plastic bonded explosive materials LX-17 and PBX 9502 (See 16th International Detonation Symposium Proceedings, 2018). Both rely on the insensitive high explosive molecule 1,3,5-triamino-2,4,6-trinitrobenzene (TATB) (Dobratz, 1995; Rice and Simpson, 1990; Boddu et al., 2010).

For LX-17 and PBX 9502 the mechanical properties are considered to be an important metric for performance and safety; therefore, the fracture properties are of significant interest. Historically the mechanical property measurements have consisted of standard techniques like compression and tensile measurements. The tensile tests are typically performed at colder temperatures and only provide an indirect, qualitative measure of fracture resistance. Also, the relatively large volume of material in the dogbone shaped tensile sample increases the probability that a randomly located impurity could nucleate a crack. Three point

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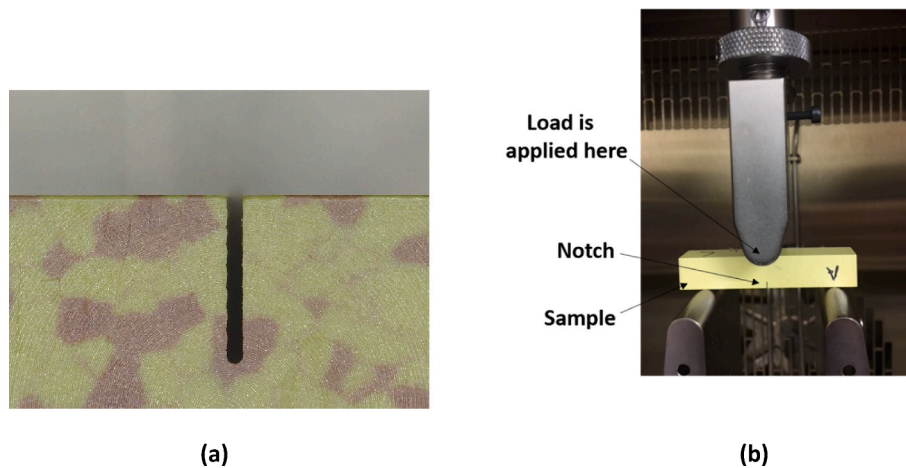


Fig. 1. (a) A 3 mm long notch in LX-17 as measured on a Keyence, (b) Photo of the experimental setup for three point bend testing of a high explosives sample.

bend measurements are a well recognized technique (Standard Test Method, 2007; Brown and Sawley, 1966; Fracture Toughness, xxxx; Williams, 2001) for measuring the fracture toughness of materials. Since an impurity would need to near the notch or along the small volume it encounters as the crack propagates through the solid, to affect the measurement, the technique is less impacted by random impurities than tensile testing. Also, unlike for tensile measurements, there exist well established analytical expressions for obtaining the fracture toughness from three point bend measurements. The fracture toughness can be important to explosive design because it allows engineers/scientists to create high margin, high confidence designs for all environments.

There have been several experimental and simulation-based studies which reported fracture results of PBX 9502 from diametral compression measurements (Palmer et al., 1993; Liu and Thompson, 2014; Liu et al., 2019; Liu et al., 2018; Liu et al., 2018; Xicheng et al., 2018; Chen et al., 2019) however, we are unaware of any studies employing this technique to LX-17. The diametral compression technique has also been applied to mock and different HE materials (Chen et al., 2006; Liu et al., 2009; Liu et al., 2010; Yuan et al., 2012; Zeng et al., 2019; Ferranti et al., 2010). The popularity of the diametral compression technique is likely because: (i) the technique utilizes small, thin disk shaped samples which require small amounts of IHE (typically ~ 1.5 g IHE) compared to rectangular three point bend specimens (typically 10.5 g IHE) and dogbone shaped tensile samples (typically 27.4 g IHE), and (ii) one can simply die press the brazil disk samples for diametral compression while three point bend and tensile IHE samples require billet pressing and the machining of samples from the billet, which require more specialized facilities due primarily to safety considerations. In diametral compression the samples simultaneously undergo compression and tension and the stress state of the sample can be rather complex. Three point bend measurements provide complimentary data to the diametral compression results and have the advantage of being simpler with potentially less interpretation/analysis error. To our knowledge there has been only one study using three point bend techniques that examined IHE's. Ferranti et al. (2010) measured LX-17 at room temperature and a single displacement rate. We have reproduced the conditions Ferranti et al used and extended the parameter testing parameter space to include a wide range of temperatures as well as a different displacement rate. We have also conducted, to our knowledge, the first three point bend fracture measurements on PBX 9502 and will report results similar to the measurement conditions performed on LX-17, enabling a direct comparison of the fracture toughness and fracture behavior of the two IHE's. And finally we shall show that under lower displacement rates one can exceed the peak load in the samples without failure occurring, enabling the x-ray CT imaging of crack propagation in the materials.

2. Experimental procedure

LX-17-1 has a wet-aminated TATB molecule and PBX 9502 has a dry-aminated one (Skidmore et al., 2011). Also the polymeric binder percentage differs slightly with LX-17 consisting of 7.5% binder and 92.5% TATB and PBX 9502 consisting of 5% binder and 95% TATB (Tarver, 2012). In this study we will be using the term 'LX-17' for simplicity but we note that we are technically using LX-17-1 (which uses wet aminated TATB). The explosive polymeric binder is usually Kel-F 800 or a more modern and almost identical version of this called FK-800. The TATB molecules are produced in the form of small particles that are subsequently mixed with a polymeric binder in a formulation process; at the end of this process roundish prill beads are produced which consist of agglomerated TATB particles with a polymeric binder that tends to be more concentrated more toward the outer periphery of the prill (United States Government Accountability Office, 2019). The prills are pressed in billets using established techniques that involve pressures up to tens of kpsi and mildly elevated temperatures in the vicinity of 100 C (Olinger, 2005; Thompson et al., 2010). Rectangularly shaped three point bend specimens are machined out of the billet to dimensions of 1.0 cm wide by 1.0 cm thick by 10.0 cm long. Sample dimensions for every sample was measured with precision calipers and micrometers. Samples are notched with a starter crack using a metal wire saw, the wire diameter was 0.22 mm and it was impregnated with diamond to bring the sample diameter to 0.26. Periodic measurements of the notch width on a commercial Keyence measurement system were consistently 0.27 mm. The notch depths were varied from 0.3 mm to 4.0 mm and the lengths of the notches of every sample was measured on the same Keyence system, a typical image is shown in Fig. 1a. Some samples were also tested without any notching and those are listed as having a 0 nm notch depth. Three point bend measurements were performed on a commercially available Instron Electropuls E3000. The displacement rate was 10^{-4} in./sec for most of the measurements but several samples were measured at a lower displacement rate of 5×10^{-6} in./sec, as noted in the text. A photo of a typical 3-point bend specimen in the testing apparatus is shown in Fig. 1b. For the LX-17 we used a single lot for all the measurements and we measured temperatures at -60 C, -20 C, 23 C, 50 C, with a few samples as high as 75 C; this was made possible by an environmental chamber attached to the Instron instrument. For the PBX 9502 samples we used three different lots and did measurements at -20 C and 23 C.

We also note that some comparison are made later in the paper to previous tensile testing results for PBX 9502. Those results were obtained on a commercially available Instron instrument on standard 3" long dogbone shaped samples at a temperature of -20 C and a strain rate of 1.67×10^{-6} /s.

The peak load is the maximum the force that the sample sustains

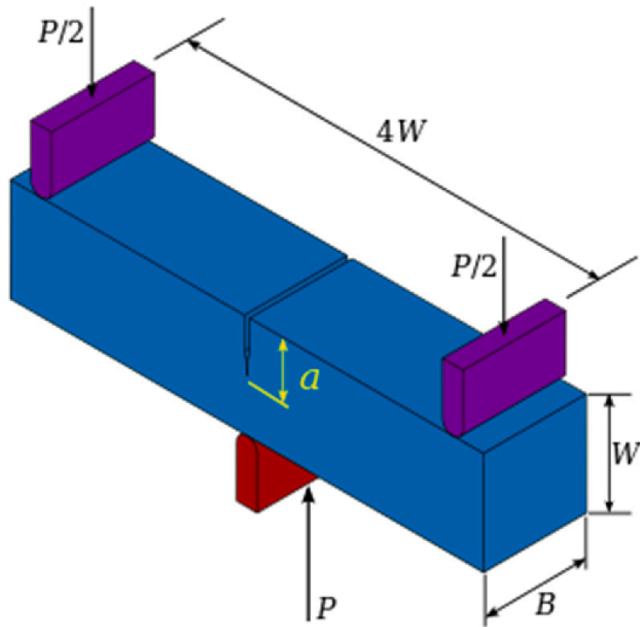


Fig. 2. Schematic illustrating the physical parameters used to convert three point bend data to fracture toughness data.

during 3-point bend testing. The peak load and the physical parameters of the samples can be used to calculate/estimate the fracture toughness; a schematic of the physical parameters of interest to are shown in Fig. 2 below. We note that a fracture toughness cannot be computed for the no notch (0 mm notch depth) samples we tested in this study and so that data is shown on other plots such as the peak load vs notch depth.

The ASTM E1290-08 and D-5045-14 standards suggest this relation (E1290: Standard Test Method, 2008; ASTM, 2014; Three-point flexural test, Wikipedia, 2021):

$$K_I = \frac{6P}{BW} Y \sqrt{a}$$

where P is the applied load and B is the sample thickness, a is the crack (notch) length, and W is the sample width, as denoted in Fig. 2. Also the factor Y is:

$$Y = \frac{1.99 - a/w(1 - \frac{a}{w})[2.15 - \frac{3.93a}{w} + 2.7(\frac{a}{w})^2]}{(1 + \frac{2a}{W})(1 - \frac{a}{W})^{3/2}}$$

Now if one substitutes the peak load or load at failure for P one can obtain K_{IC} , the fracture toughness.

For the PBX 9502 samples we had three different lots available which

we will arbitrarily call lots A, B, and C. Lot A is an older lot and lots B and C are newer (past 5 years).

X-ray CT scans were performed on a commercially available North Star Imaging (NSI) model X25 instrument; the x-ray source was operated at 50 kV and 100 μ A.

3. Results and discussion

3.1. Temperature and notch depth dependence

3.1.1. LX-17

Fig. 3a shows a typical load vs time plot for a 3-point bend test. The peak load (to failure) values for all of the tests were extracted and Fig. 3b shows the peak load versus temperature for temperatures of -60 C to 50 C for samples with notch depths of 0 (not notch), 0.3 mm, and 1.0 mm. At colder temperatures there is a significantly greater difference in the peak load between the notched and unnotched samples than between the (0.3 mm and 1.0 mm) notched samples. This is likely due to the greater force/energy needed to nucleate the crack for the no notch sample. With warmer temperatures the peak load difference between the notched and unnotched samples decreases, likely because as the temperature approaches and exceeds the glass transition temperature (T_g) of the polymeric binder, which is approximately 27 C, the overall strength of the LX-17 decreases significantly and less force/energy is needed to nucleate a crack (Groves and Cunningham, 2000).

Fig. 4a shows the peak load for the same temperatures but as a function of the notch depth (for nonzero notch depths). The peak load generally decreases with increasing notch depth as expected. The data shown in Fig. 4a was used to calculate the fracture toughness for different temperatures as a function of the notch depth, as shown in Fig. 4b. The fact that the fracture toughness is significantly lower at 50 C, and higher at -60 C, is not surprising, what is a little unexpected is that the fracture toughness values at -20 C and 23 C are as close as they are.

Similar to what was noted by Ferranti et al. (2010) for ambient temperature (23 C) LX-17 measurements, we observe the fracture toughness being lower for notch depths < 1 mm and being relatively constant for notch depths ≥ 1 mm. At 23 C we observe an average value of 0.427 ± 0.025 MPa(m) $^{0.5}$ for the fracture toughness which is relatively near the 0.366 ± 0.019 MPa(m) $^{0.5}$ average value reported by Ferranti et al at 23 C. We have no literature values to compare the values at other temperatures because there are no literature values (i.e. the first values are appearing in this paper). The fracture toughness of these explosive materials is relatively low; for example, a Titanium alloy, a SiC

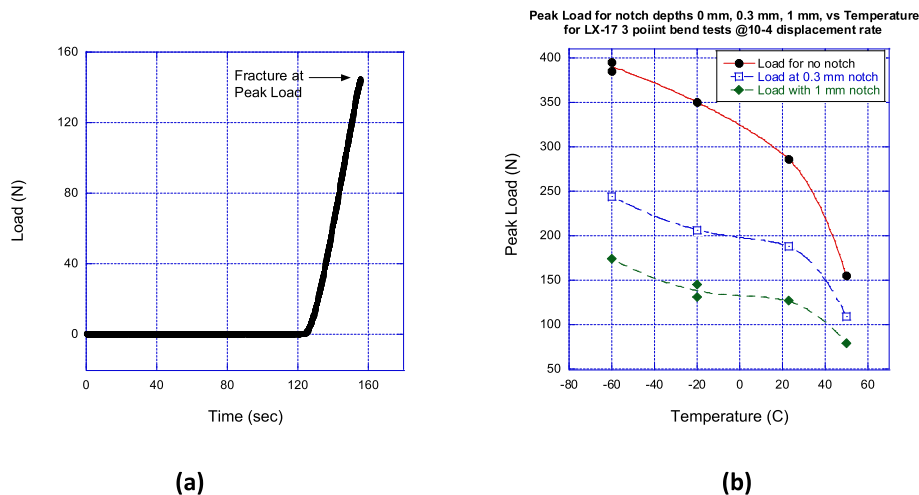


Fig. 3. 3-Point bend measurements on LX-17: (a) Individual load vs time plot at $T = -20$ C and (b) Peak load vs temperature for samples with notch depths of 0 (no notch), 0.3, and 1.0 mm.

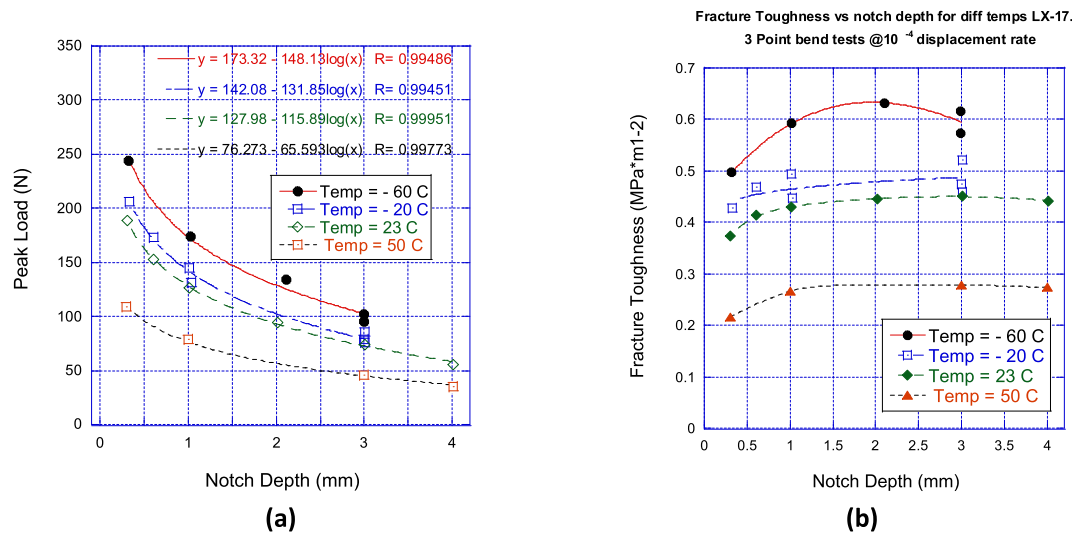


Fig. 4. LX-17 3-point bend measurements on samples as a function of notch depth over a wide temperature range of -60 C to 50 C: (a) Peak load and (b) Fracture toughness calculated from the data shown in (a) and the measured physical parameters of the samples (including the notch depth).

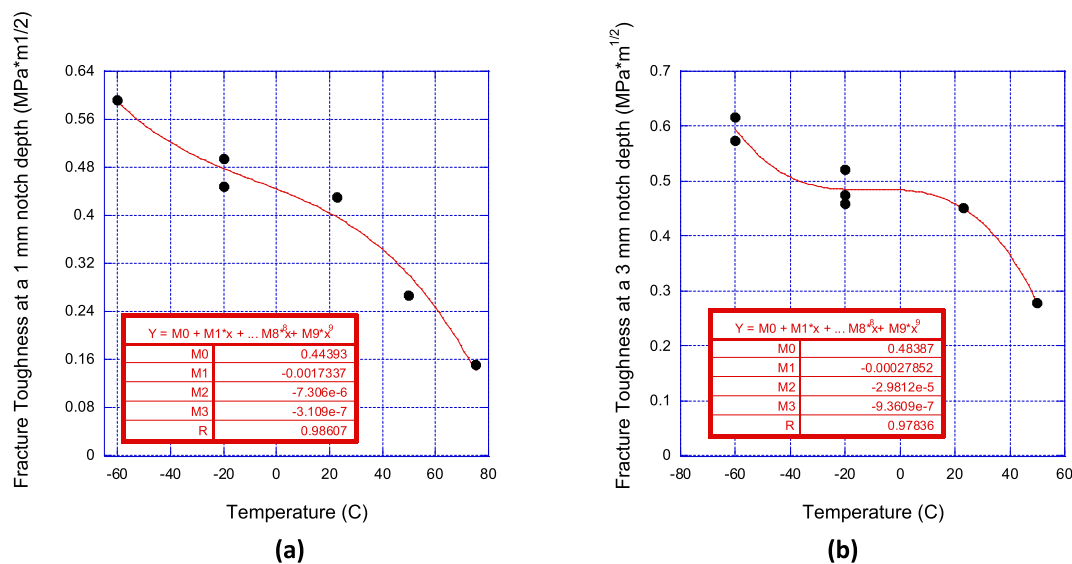


Fig. 5. Fracture toughness vs Temperature for LX-17 samples with a fixed notch depth of (a) 1 mm and (b) 3 mm.

ceramic, and PMMA have fracture toughness values of approximately > 90 , $3-4$, and $0.7-1.6$ MPa(m)^{0.5} respectively (Hertzberg, 1989).

The fracture toughness as a function of temperature for two different notch depths, 1 mm and 3 mm, are shown in Fig. 5 a and b respectively. In Fig. 5a data was obtained over an extra wide temperature range of -60 C to 75 C while in Fig. 5b data was obtained from -60 C to 50 C. While a reasonable linear fit can be obtained for both data sets, a third order polynomial presents the best fit for both sets of data, likely due to the fact that the fracture toughness does not change as much in the region from -20 C to 23 C than it does as the temperature goes significantly below -20 C or above 23 C.

3.1.2. PBX 9502

Before we undertook this study we there were in-house measurements from four years earlier of two different PBX 9502 lots (lots D and E) that were not analyzed or documented. We located the peak load vs notch depth data from then, as well as pristine samples from those same lots that were never notched nor tested. Before beginning our measurements of our three PBX 9502 lots of interest (i.e., lots A, B, and C) we notched and performed three-point bend measurements on several lot D

and E samples. In Fig. 6 below we show the peak load values at different notch depths for samples measured recently and measured 4 years earlier at ambient temperature (23 C). The points generally overlap to a remarkable degree, suggesting that the three-point measurements on the same instrument on samples from the same lots are quite reproducible using this technique even after significant periods of time.

In Figs. 7 and 8 the peak load and fracture toughness are shown as a function of notch depth for the three different lots A, B, and C, at temperatures of -20 C and 23 C. At a testing temperature of 23 C averaged over 12 measured values/samples for each lot, the fracture toughness is 0.432 ± 0.036 , 0.337 ± 0.031 , 0.402 ± 0.031 MPa(m)^{0.5}, for lots A, B, and C respectively. We note that these values are generally near the 0.427 MPa(m)^{0.5} fracture toughness value we observed at 23 C for LX-17 reported earlier in this paper. In terms of comparing PBX 9502 fracture toughness values with the literature, Liu et al (Liu et al., 2018) estimated a value of $\sim 0.58 - 0.67$ MPa(m)^{0.5} at ambient temperatures which is reasonably close if one considers that they used a completely different technique with different conditions on different form factor samples.

In Table 1 below the average percent difference between the fracture toughness values at -20 C and 23 C for each lot is listed. The difference

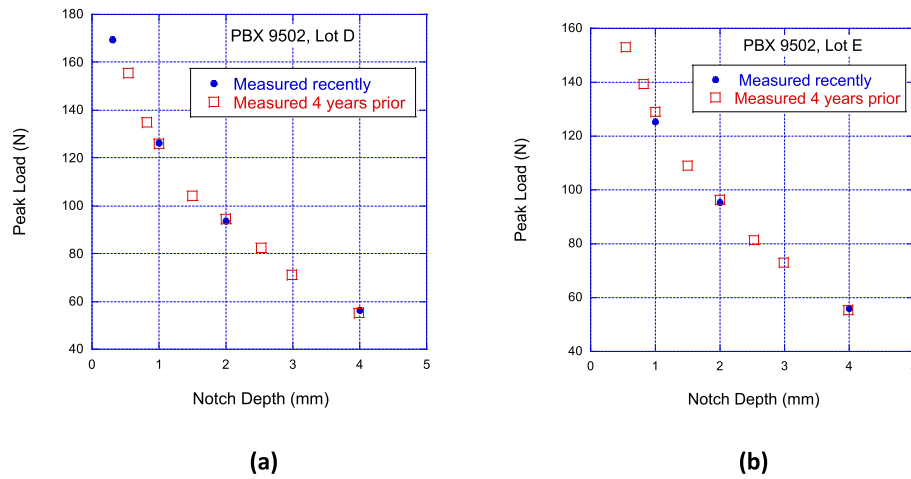


Fig. 6. Peak load as a function of notch depth for PBX 9502 (a) lot D and (b) lot E samples. Samples from these lots were measured recently and 4 years prior and suggest that the reproducibility over time with the technique is excellent.

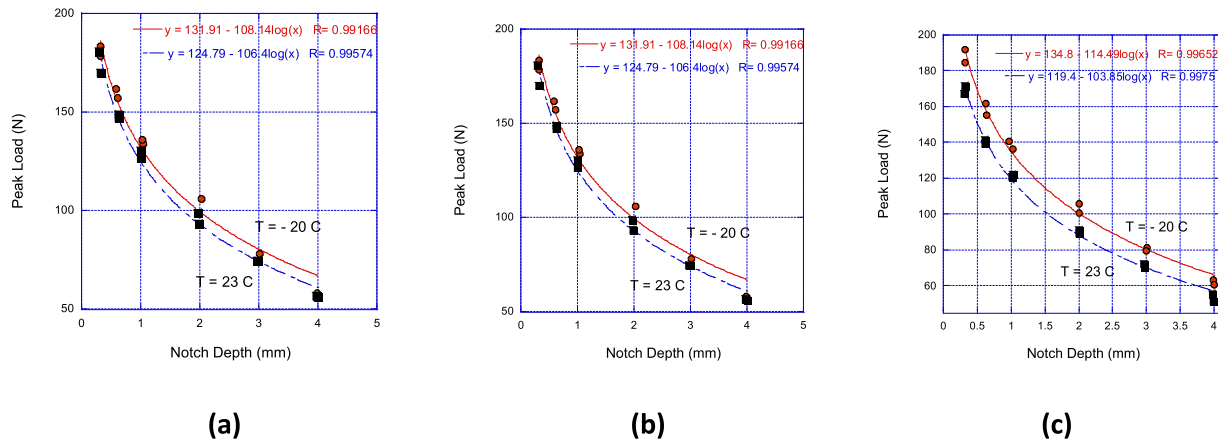


Fig. 7. Peak load vs notch depth for PBX 9502 3-point bend measurements at $T = -20\text{ C}$ and 23 C for (a) lot A, (b) lot B, (c) lot C.

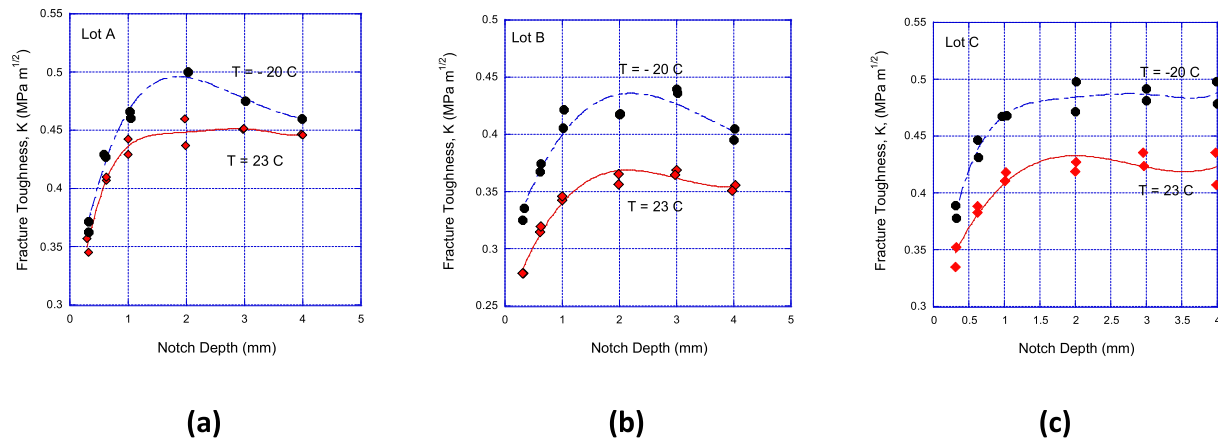


Fig. 8. Fracture toughness vs notch depth for PBX 9502 3-point bend measurements at $T = -20\text{ C}$ and 23 C for (a) lot A, (b) lot B, (c) lot C.

is the smallest ($\sim 5.2\%$) for lot A which has the highest fracture toughness at ambient and is the largest ($\sim 16\%$) for lot B which has the lowest fracture toughness at ambient.

As noted in the introduction oftentimes tensile testing data is taken as a proxy for fracture toughness. We have tensile data on samples from lots A, B, C, at -20 C , and so a comparison is possible. Fig. 9a shows the fracture toughness vs notch depth for lots A,B,C, at $T = -20\text{ C}$ and Fig. 9b

shows the fracture toughness vs. the measured tensile strength for samples from the same lots. In Fig. 9b if one compares the lot B results from the combined lots A and C results, which show a significant difference, there is a correlation of higher fracture toughness with higher tensile strength. However, if one compares the lot A and lot C results in Fig. 9, where the differences in values are much smaller, a trend of fracture toughness with tensile strength is less clear, where lot A has a

Table 1

Mean % difference between fracture toughness values for each PBX 9502 lot.

PBX 9502 Lot	Mean % difference between fracture toughness values at -20 C and 23 C	Standard deviation
A	5.2%	1.6%
B	16.0%	3.5%
C	12.8%	2.5%

slightly lower fracture toughness with a higher tensile strength.

3.2. Crack propagation in LX-17 and PBX 9502

The results discussed to this point have all been at a displacement rate of 10^{-4} in/sec. We also measured several samples at a much lower displacement rate of 5×10^{-6} in/sec and in Fig. 10 below we show the fracture toughness for PBX 9502 lot A samples with notch depths of 3 and 4 mm at the two displacement rates. We observed relatively small differences in the fracture toughness as a function of displacement rate because the peak load values that we measured were not a strong function of the displacement rate over that range. However, we did observe a dramatic effect of the displacement rate when it came to sample failure.

Fig. 11 below shows load vs time plots for an (a) LX-17 and (b) PBX 9502 sample in which the displacement rate was the lower 5×10^{-6} in/s value. In contrast to our standard displacement rate (Fig. 3 a shown before) where there is catastrophic failure (ie a splitting of the sample and inability to hold the load) after the peak load, the data in Fig. 11 shows the load decreases gradually after the peak load is reached until the measurement was ended. Visual observations during and after the testing confirmed that the samples remained completely intact when the lower displacement rate was employed. This is fortuitous since to understand crack propagation in these materials it is useful to be able to see (image) crack propagation and when a sample catastrophically breaks/fails it becomes difficult if not impossible to do this with sufficient accuracy. The intact samples obtained at the lower displacement rate enabled us to image the path of crack propagation relative to the microstructure within a CT x-ray tool. We also note that the peak loads for the LX-17 and PBX 9502 data shown in Fig. 11 were similar with 72.2 N for the LX-17 and 68.2 N for the PBX 9502. We also note in Fig. 11 that the applied load varied around it's peak, in contrast to the more abrupt drop in load after the peak is reached consistent with brittle materials. This suggests that at least under some conditions these high explosive materials exhibit more *quasi-brittle* behavior which is consistent with work by Liu and Thompson (2014), Liu et al. (2018); Liu et al. (2010).

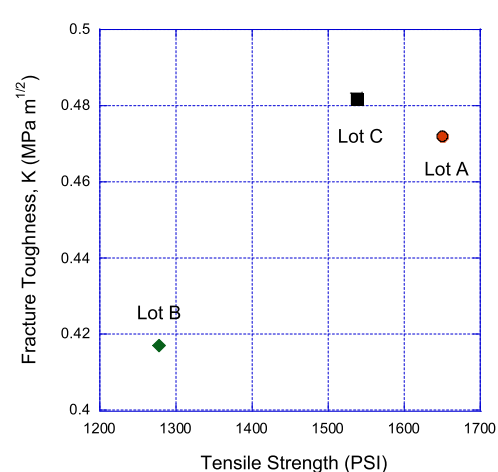
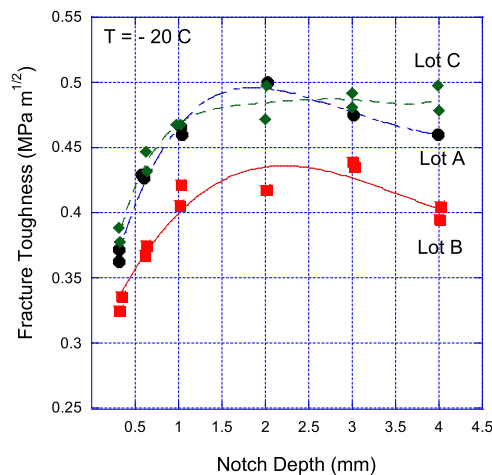


Fig. 9. Results at T = -20 C for PBX 9502 lots A,B,C for the fracture toughness vs (a) notch depth and (b) tensile strength.

Fig. 12 below shows x-ray CT images of the LX-17 sample tested at the lower displacement rate. Fig. 12a is a lower magnification image showing the full notch and the crack which dissipates toward the end and Fig. 12b is a higher magnification image from a different CT slice. In the images one can see the outline of the prills that are pressed to form the material. Note how the crack tends to go in a straight line from the notch with little change in direction from the prills and the microstructure. In addition to the images shown in Fig. 12 there are many CT slices not shown which have been examined and which consistently show the same behavior.

Fig. 13 shows x-ray CT images from the two PBX 9502 samples tested at the lower displacement rate. (a) and (b) are lower magnification images from each sample showing the full notch and the crack which dissipates toward the end and (c) is a higher magnification image from a different CT slice in the second PBX 9502 samples. Note how unlike the LX-17 images shown in Fig. 12, the cracks in the PBX 9502 show changes in direction. In addition to the images shown in Fig. 13 there are many CT slices not shown which have been examined and which consistently show the same behavior. It appears from this data that crack propagation is affected more by microstructural features in the PBX 9502 than in

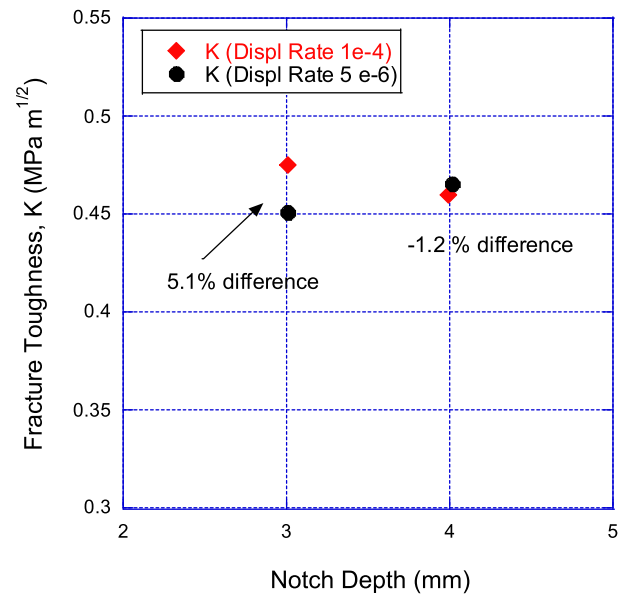


Fig. 10. Fracture toughness vs notch depth at T = -20 C for PBX 9502 lot A samples tested at two different displacement rates, 10^{-4} and 5×10^{-6} in/sec.

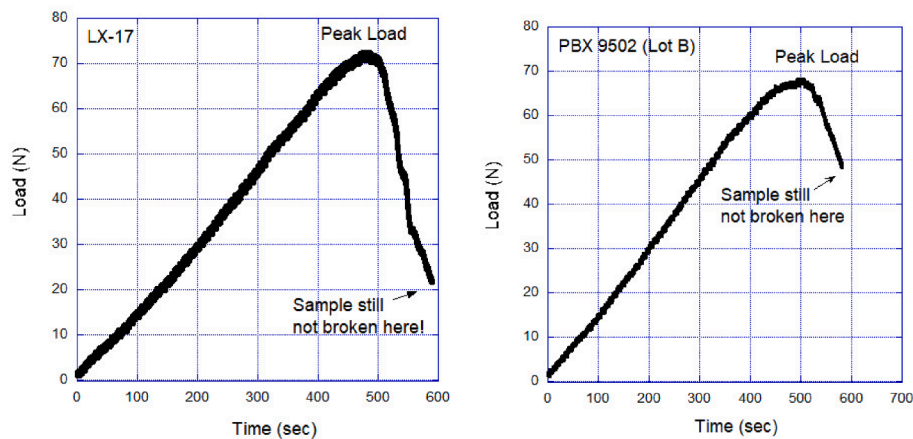


Fig. 11. Three-point bend load vs time at $T = -20\text{ }^{\circ}\text{C}$ for a low displacement rate ($5 \times 10^{-6}\text{ mm/s}$) for (a) LX-17 and (b) PBX 9502 lot B samples. The time at which the curve ends is due to the ending of the test and not due to sample failure. The notch depths were 3 mm for these samples.

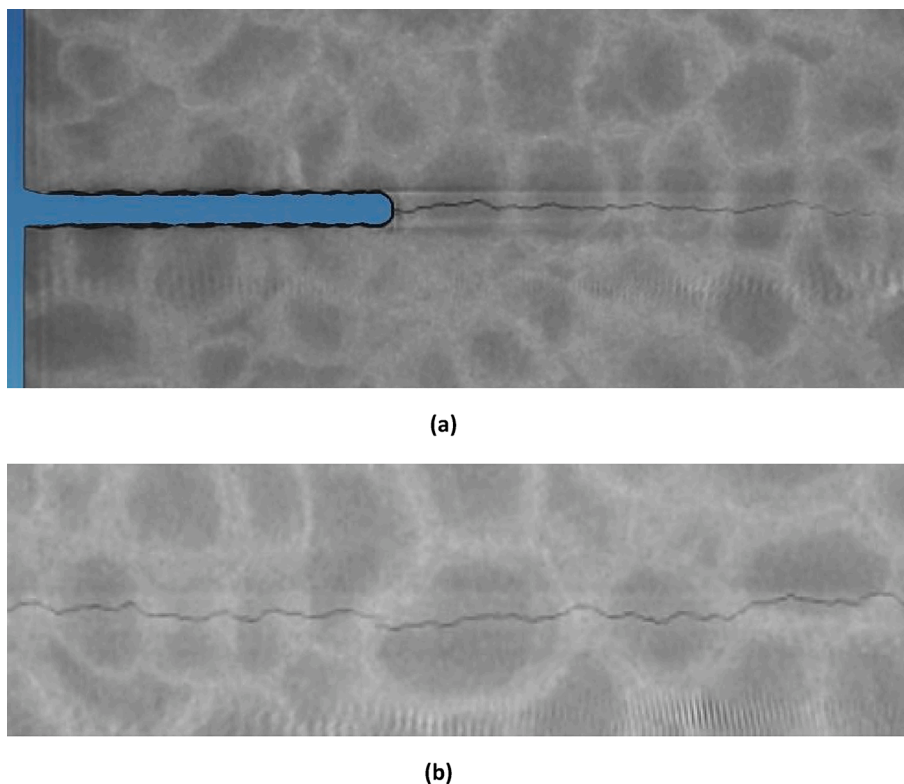


Fig. 12. X-ray CT image of the LX-17 sample tested at the lower displacement rate. (a) is a lower magnification image showing the full notch and the crack which dissipates toward the end and (b) is a higher magnification image from a different CT slice.

the LX-17. While there can be differences between TATB lots in terms of particle size and distribution, roughness, etc., that can influence properties such as the compressive strength (Mirkarimi et al., 2018), it is primarily the % binder (5% for PBX 9502 vs 7.5% for LX-17) and the fact that PBX-9502 uses dry-aminated TATB vs. wet-aminated TATB for LX-17, that differentiates the two IHE's.

Why would the LX-17 respond differently than PBX 9502 to crack propagation? There are two notable differences between the two HE's. The first is that LX-17 has about 7.5% binder vs 5% for PBX 9502. If the cracks preferentially propagate in the binder rich areas of PBX 9502 (i.e. at prill boundaries) then since LX-17 has even more binder, one might expect that the cracks might also prefer to propagate in the binder rich areas of LX-17, which is not what is observed. The second notable difference is that LX-17 uses wet-aminated TATB and PBX 9502 uses dry-

aminated TATB. We are unaware of any mechanical property studies looking at wet-aminated vs dry-aminated TATB (with the binder % held constant) but this work suggests that cracks may have a greater tendency to penetrate wet aminated TATB than dry aminated TATB. We note that the fracture toughness values for the LX-17 and PBX 9502 are similar so the greater deflection of the cracks in the PBX 9502 microstructure did not manifest itself significantly in the fracture toughness; likely compensating factors arising from the differences between the two materials (different lots, binder %, etc) come into play.

We note that there is potentially an effect of confinement, i.e., less confinement near the surface compared to the interior, on the crack profile. However, since the crack profile is impacted by the microstructure which can change with depth into the solid, it's not possible/straightforward to determine if greater confinement has an impact on

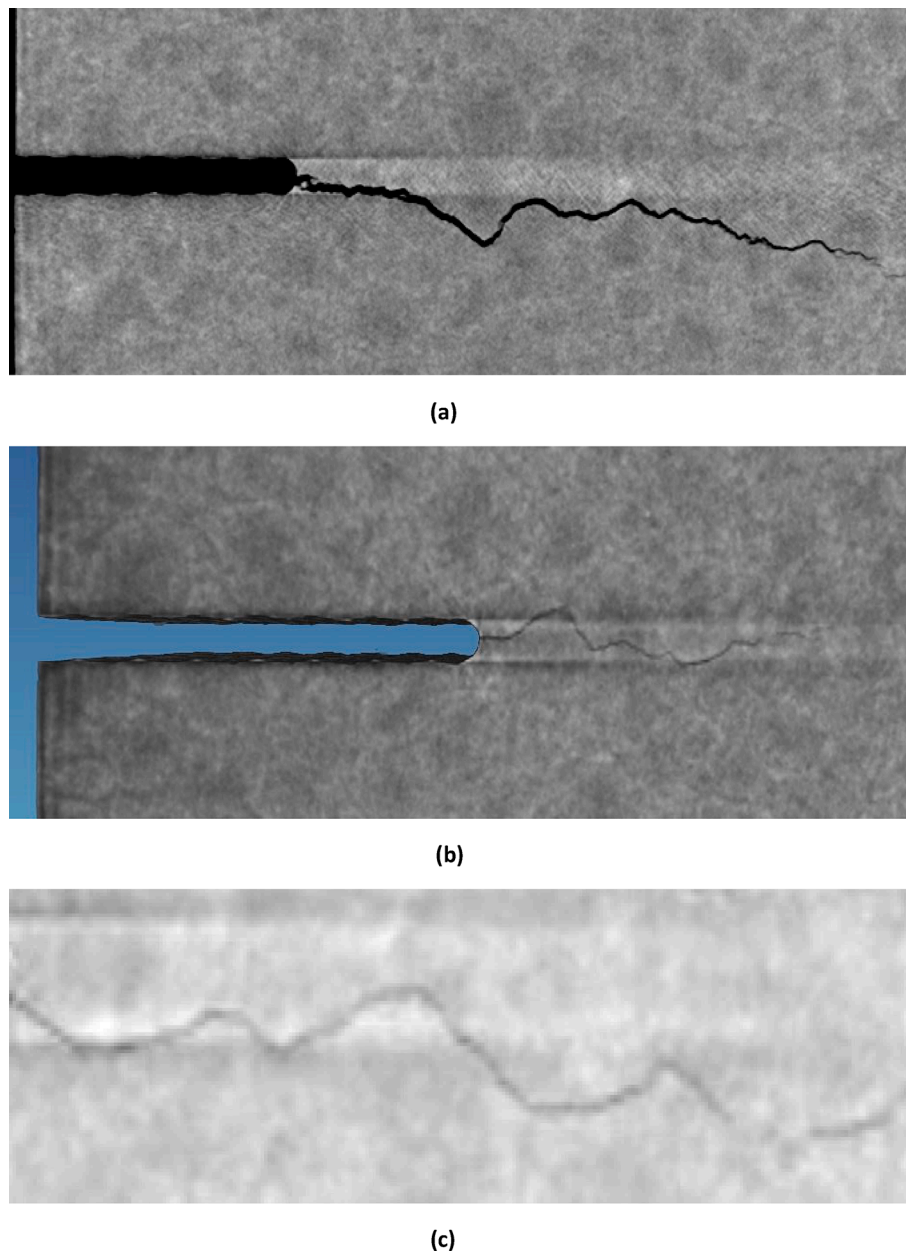


Fig. 13. X-ray CT images from the two PBX 9502 samples tested at the lower displacement rate. (a) and (b) are lower magnification images from each sample showing the full notch and the crack which dissipates toward the end and (c) is a higher magnification image from a different CT slice in the second PBX 9502 samples.

the crack profile.

4. Summary

In this work we present, for the first time, a study of fracture as a function of temperature and displacement rate and starter crack length on billet pressed IHE materials using a standard three point bend technique. We demonstrated that the technique can yield consistent results over time by measuring samples from the same lots 4 years later. We observed that the fracture toughness of LX-17 and PBX 9502 are similar; for example at 23 C and a 10^{-4} in/s displacement rate we observed $0.427 \text{ MPa(m)}^{0.5}$ and $0.337\text{--}0.432 \text{ MPa(m)}^{0.5}$ for LX-17 (single lot) and PBX 9502 (3 different lots) respectively. We also determined the temperature dependence for different starter crack lengths (notch depths) and found that the fracture toughness generally decreases with larger starter crack sizes and temperatures. For the LX-17, where a wide

temperature range of -60 C to 50 C was investigated, the impact of temperature on fracture toughness between -20 C and 23 C was less than expected. For the PBX 9502 we observed the difference in fracture toughness values between -20 C and 23 C did depend on the lot. We had tensile test data on the three lots and the data suggested a general trend of increasing tensile strength with increasing fracture toughness.

We also investigated the effect of a lower displacement rate of 5×10^{-6} in/sec and observed that while the impact on the fracture toughness was relatively small, the samples surprisingly did not break (fail) after exceeding the peak load. We took advantage of this to perform x-ray CT imaging of the samples to see crack propagation in the intact samples. We observed that cracks were generally straight and unaffected by the microstructure in the LX-17, it passed directly through the remnant prills visible in the CT scans. This is in contrast to PBX 9502 where the direction of the cracks would frequently change and were likely influenced by the underlying microstructure.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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